Master of Science in Information (MSI) Degree Requirements

Fall 2025 Cohort - Big Data Analytics

Graduation Requirements

Additional details and academic policies are located in the MSI and MHI Student Handbook.

Credits & GPA	Courses
• **Credits:** ○ 39 SI credits ○ 48 total graduate-level credits (500-level or higher) • **GPA:** ○ Minimum cumulative **3.0 GPA**	• **Completion of:** ○ Core Courses ○ Mastery Prerequisite Courses ○ Mastery Course ○ Selective Requirements “Pick 2” ○ Elective Courses • Minimum of **C-** grade in all courses

Important Terminology

Course Types	Acronyms
• **Prerequisite** (prereq): a requirement to complete **before** the listed course. • **Corequisite** (coreq): a requirement to complete **before or during** the same semester as the listed course.	• **Semester Types:** ○ F = Fall semester ○ W = Winter semester • **Courses:** ○ SI = School of Information

Core Courses

Core Courses should be taken during the first semester of the MSI degree program.

Course	Credits	Semester
☐ **SI 500:** Problem-Solving with People, Information, and Technology	3	F
☐ **SI 506:** Programming I	3	F / W

Mastery Prerequisite Courses

The Mastery Prerequisite Courses are a set of mandatory courses that must be completed to meet the

eligibility requirements for enrolling in the Mastery Course (SI 699).

Course	Credits	Semester
☐ **SI 504:** Servers, The Shell, and Git	1.5	F / W
☐ **SI 507:** Intermediate Programming ***Prereq:*** *SI 506* | ***Coreq:*** *SI 504*	3	F / W
☐ **SI 544:** Introduction to Statistics and Data Analysis	3	F
☐ **SI 568:** Becoming a Data Scientist ***Coreq:*** *SI 618*	1.5	W
☐ **SI 602:** Math Foundations ***Prereq:*** *SI 504, SI 506, SI 544*	3	F / W
☐ **SI 618:** Data Manipulation and Analysis ***Prereq:*** *SI 504, SI 506* | ***Coreq:*** *SI 507*	3	F / W
PICK ONE (Selective Requirement) ☐ **SI 670:** Applied Machine Learning ***Prereq:*** *SI 507, SI 544, SI 618* | ***Coreq:*** *SI 602* ☐ **SI 671:** Data Mining ***Prereq:*** *SI 507, SI 544, SI 618* | ***Coreq:*** *SI 602*	3 3	F F

Mastery Course

The Mastery course (SI 699) serves as the culminating capstone course for the MSI degree.

Course	Credits	Semester
☐ **SI 699:** Big Data Analytics ***Required to complete Core and Mastery Prerequisite Courses first***	3	F / W

Selective Requirement (Pick 2)

Students must complete two Selective Requirement courses from the list below, which may be taken concurrently with the Mastery Course.4

Course	Credits	Semester
☐ **SI 608:** Networks ***Prereq:*** *SI 507,* ***Coreq:*** *SI 618*	3	F
☐ **SI 630:** Natural Language Processing: Algorithms and People ***Prereq:*** *SI 506, SI 507, SI 544, & SI 618*	3	W

☐ **SI 649:** Information Visualization ***Prereq:*** *SI 506, SI 507, SI 544*	3	F / W
☐ **SI 650:** Information Retrieval ***Prereq:*** *SI 507, SI 618*	3	F
☐ **SI 670:** Applied Machine Learning ***Prereq:*** *SI 507, SI 544, SI 618* **|** ***Coreq:*** *SI 602* **AND** **SI 671:** Data Mining ***Prereq:*** *SI 507, SI 544, SI 618* **|** ***Coreq:*** *SI 602*	3 3	F F

Additional Course Planning Considerations

Course Waivers

The UMSI Course Waiver Policy is designed to allow UMSI students, through one of three methods, to demonstrate proficiency and knowledge in the subject matter of a particular course in lieu of formally completing the course. There are three potential ways to waive an SI course: Skill-Based Waivers, Individual Review Waivers, and Automatic Waivers.

- ***NOTE:*** *Course waivers do not count for credit, nor reduce the number of credits required to complete a degree. Instead, receiving a course waiver allows the student to choose a more advanced course or another course of interest to fulfill the credit requirement. More information on course waivers can be found on the UMSI Course Waivers site. Questions can be directed to umsi.waivers@umich.edu.*

Internship Course

MSI students are eligible to receive up to 9 credits for approved internships through the UMSI Internship program. Most internships are typically completed in the Summer semester between the first and second year in the program, with the credits backdated to the prior winter semester:

- **SI 681:** Internship/Field Experience *(most common option for summer internships)*
- **SI 690**: Internship/Field Experience *(most common for F / W internships)*

For more information about the internship, refer to the MSI Internship Program Handbook.

Advised Course Sequence

The advised course sequence represents the courses and the order that certain courses should be taken in each term to promote student success and degree completion. It will provide clarity by identifying an efficient, timely path for completing the **Big Data Analytics** pathway.

It is important to note that the advised course sequence is not an official degree plan, but rather serves as a tool for proactive discussion to plan for required and elective courses with an MSI/MHI Academic Advisor. Students should confirm their course selections with an Academic Advisor to ensure a smooth registration process and timely degree completion in two academic years (four terms). Students can visit the MSI Academic Advising Resource site to learn how to connect with an Academic Advisor.

The following document contains the advised course sequence:
Course Planning Worksheet for Big Data Analytics (Google Sheet)